

5.3 PS9400

5.3.1 General information

The supply module PS9400 is used together with an X20 bus controller. It is equipped with a feed for the bus controller, the X2X Link, and the internal I/O supply.

- Supply for the bus controller, X2X Link, and internal I/O supply
- Feed and bus controller / X2X Link supply electrically isolated
- Redundancy of bus controller / X2X Link supply possible by operating multiple supply modules simultaneously

5.3.2 Order data

Model number	Short description	
	Supply modules	
X20PS9400	X20 supply module for bus controller and internal I/O supply, X2X link bus supply	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BB80	Basic X20 bus controller, for bus controller fieldbus interface and bus controller supply module, X20 connection	

Table 54: PS9400 - order data

5.3.3 Technical data

Product ID	PS9400
Short description	
Power supply module	24 VDC supply module for bus controller, X2X Link bus supply and I/O
Bus controller / X2X Link supply input	
Input voltage	24 VDC (-15 % / +20 %)
Input current	Max. 0.7 A
Reverse polarity protection	Yes
Fuse	Integrated, cannot be exchanged
Bus controller / X2X Link supply output	
Rated output power	7.0 W
Parallel operation	Yes ¹⁾
Redundant operation of bus controller / X2X Link supply	Yes
I/O supply input	
Input voltage	24 VDC (-15 % / +20 %)
Fuse	Recommended pre-fusing max. 10 A slow-blow
I/O supply output	
Rated output voltage	24 VDC
Allowable contact load	10.0 A
General information	
Status indicators	Overload, operating status, module status
Diagnostics Module run error Overload	Yes, with status LED and software status Yes, with status LED and software status
Electrical isolation Bus controller / X2X bus supply I/O supply	Yes No
Certification	CE, C-UL-US (in development), GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20

Table 55: PS9400 - technical data

Product ID	PS9400
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order bus controller base 1 x X20BB80 separately

Table 55: PS9400 - technical data (cont.)

- 1) In parallel operation, only 75 % of the rated power can be assumed. Please ensure that all parallel operating power supplies are switched on and off simultaneously.

5.3.4 Additional technical data

Product ID	PS9400
Bus controller / X2X Link supply output	
Overload behavior	Short circuit / temporary overload protection
I/O supply input	
Reverse polarity protection	No
I/O supply output	
Behavior at short circuit	Recommended protection
General information	
B&R ID code	\$1F8C

Table 56: PS9400 - additional technical data

5.3.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Double flash	Indicates one of the following conditions: - The bus controller / X2X Link power supply is overloaded - I/O supply too low - Input voltage for bus controller / X2X Link too low
	e + r	Steady red / single green flash		Invalid firmware
	l	Red	Off	The bus controller / X2X link supply is within the acceptable range
			On	The bus controller / X2X Link power supply is overloaded

Table 57: PS9400 - status indicators

5.3.6 Pin assignments

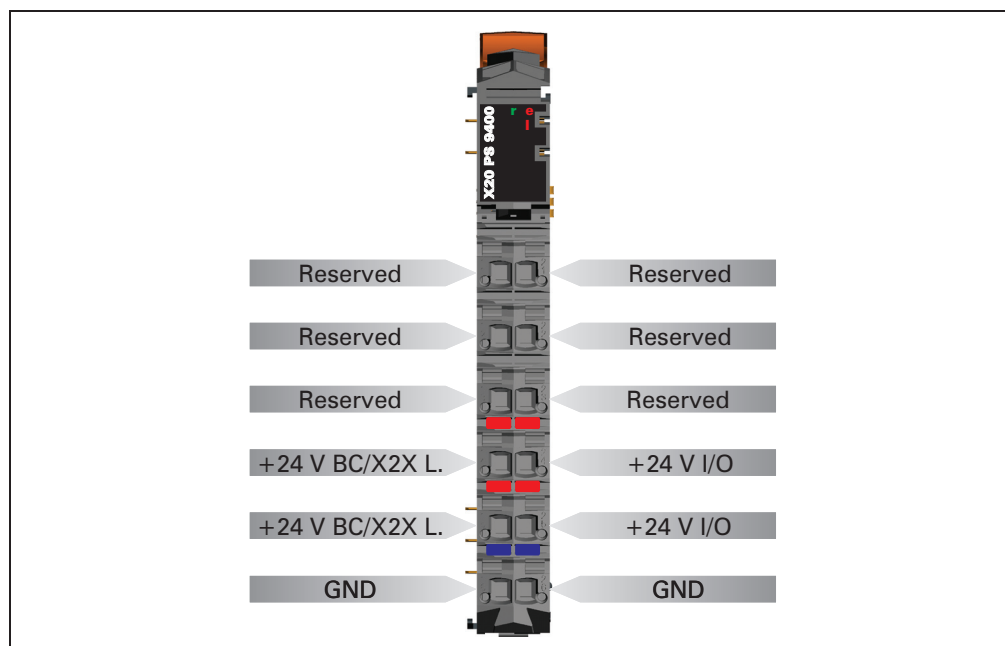


Figure 40: PS9400 - pin assignments

5.3.7 Connection examples

With two separate supplies

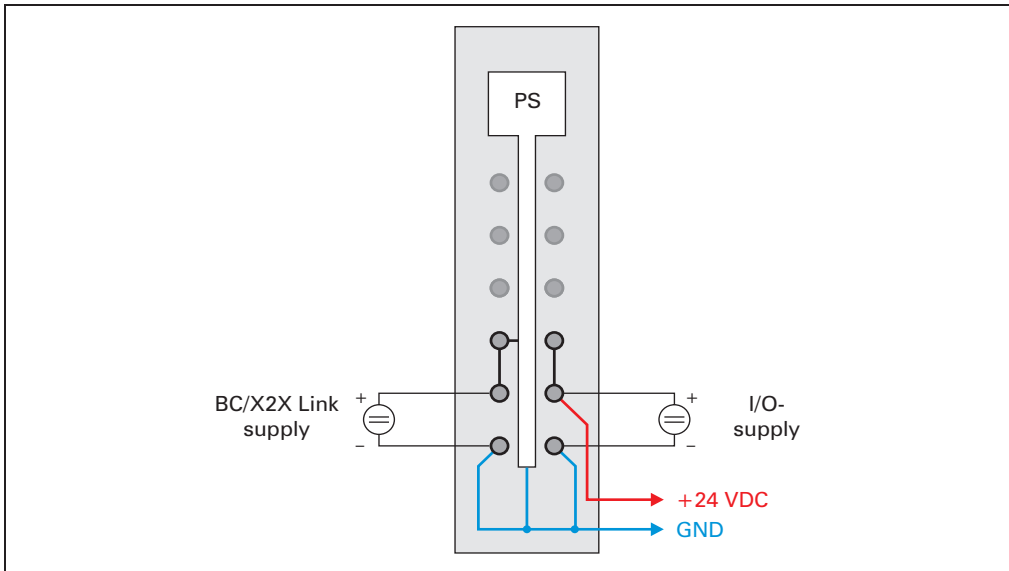


Figure 41: PS9400 - connection example with two separate supplies

With a supply and jumper

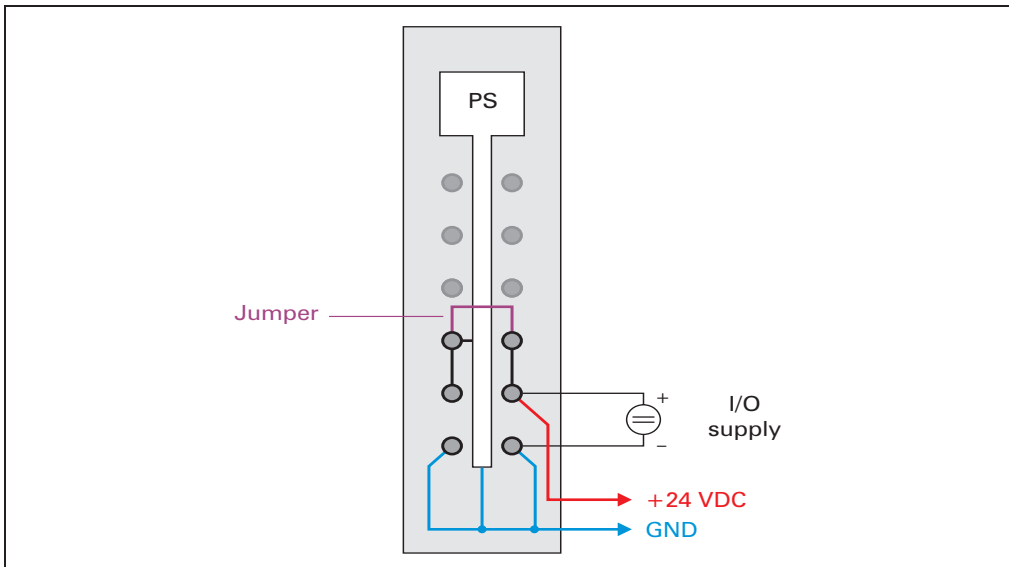


Figure 42: PS9400 - connection example with a supply and jumper

5.3.8 Module status

The following module states are monitored:

- Bus supply current
- Bus supply voltage
- 24 VDC I/O supply voltage

Bus supply current

A bus supply current of >2.3 A is displayed as a warning.

Bus supply voltage

A bus supply voltage of <4.7 V is displayed as a warning.

24 VDC I/O supply voltage

An I/O supply voltage of <20.4 V is displayed as a warning.

5.3.9 Bus supply current

The bus supply current is measured with a resolution of 0.1 A.

5.3.10 Bus supply voltage

The bus supply voltage is measured with a resolution of 0.1 V.

5.3.11 B&R ID code

Code for module identification (\$1F8C).

5.3.12 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
$\geq 100 \mu\text{s}$

5.3.13 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
2 ms

6. Bus transmitters and bus receivers

6.1 General information

The bus receiver BR9300 is used to connect the X20 System directly to the remote X2X link backplane. The bus sender BT9100 is used to connect to the next station. See also chapter 2 "System characteristics", section 3 "Decentralized backplane" on page 35.

6.1.1 Overview

Model number	Short description
X20BR9300	X20 bus receiver (X2X Link) with feed for internal I/O supply, and X2X Link bus supply
X20BT9100	X20 bus transmitter (X2X Link)

Table 58: Overview of bus transmitters and bus receivers

6.2 BR9300

6.2.1 General information

The bus receiver BR9300 is used to connect the X20 system to the X2X link. The module is equipped with a feed for the X2X link as well as the internal I/O supply.

The left and right locking plates are included in the delivery (for info on installation see chapter 6 "Mechanical design", page 429).

- X2X Link bus receiver
- Supply for X2X Link and internal I/O supply
- Electrical isolation of feed and X2X Link supply
- Redundancy of X2X Link supply possible by operating multiple supply modules simultaneously

6.2.2 Order data

Model number	Short description	Image
	Bus Receiver	
X20BR9300	X20 bus receiver (X2X Link) with feed for internal I/O supply, and X2X Link bus supply	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM01	X20 supply bus module, internal I/O supply is split to the left	

Table 59: BR9300 - order data

6.2.3 Technical data

Product ID	BR9300
Short description	
Bus receiver	X2X Link bus receiver with supply for I/O and bus

Table 60: BR9300 - technical data

X20 System modules • Bus receivers and bus transmitters • BR9300

Product ID	BR9300
X2X Link supply input	
Input voltage	24 VDC (-15 % / +20 %)
Input current	Max. 0.7 A
Reverse polarity protection	Yes
Fuse	Integrated, cannot be exchanged
X2X Link supply output	
Rated output power	7.0 W
Parallel operation	Yes ¹⁾
Redundant operation	Yes
I/O supply input	
Input voltage	24 VDC (-15 % / +20 %)
Fuse	Recommended pre-fusing max. 10 A slow-blow
I/O supply output	
Rated output voltage	24 VDC
Allowable contact load	10.0 A
General information	
Status indicators	X2X bus function, overload, operating status, module status
Diagnostics Module run error X2X bus function Overload	Yes, with status LED and software status Yes, with status LED Yes, with status LED and software status
Electrical isolation X2X Link supply I/O supply	Yes No
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing

Table 60: BR9300 - technical data (cont.)

Product ID	BR9300
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order supply bus module 1 x X20BM01 separately

Table 60: BR9300 - technical data (cont.)

- 1) In parallel operation, only 75 % of the rated power can be assumed. Please ensure that all parallel operating power supplies are switched on and off simultaneously.

6.2.4 Additional technical data

Product ID	BR9300
X2X Link supply output	
Overload behavior	Short circuit / temporary overload protection
I/O supply input	
Reverse polarity protection	No
I/O supply output	
Behavior at short circuit	Recommended protection
General information	
B&R ID code	\$1BC1

Table 61: BR9300 - additional technical data

6.2.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Double flash	Indicates one of the following conditions: - X2X link power supply is overloaded - I/O supply too low - Input voltage for X2X Link supply too low
	e + r		Steady red / single green flash	Invalid firmware
	X	Orange	Off	No communication at the X2X link
			On	X2X link communication in progress
	I	Red	Off	X2X link supply in the acceptable range
			On	X2X link power supply is overloaded Solution: Use an additional feed module PS3300

Table 62: BR9300 - status indicators

6.2.6 Pin assignments

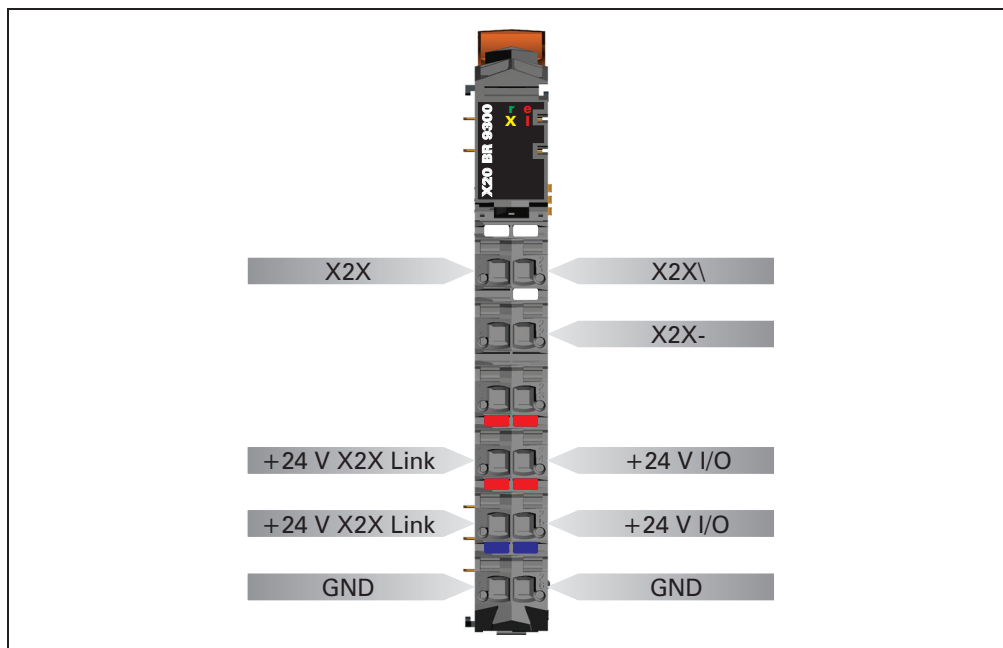


Figure 43: BR9300 - pin assignments

6.2.7 Connection examples

With two separate supplies

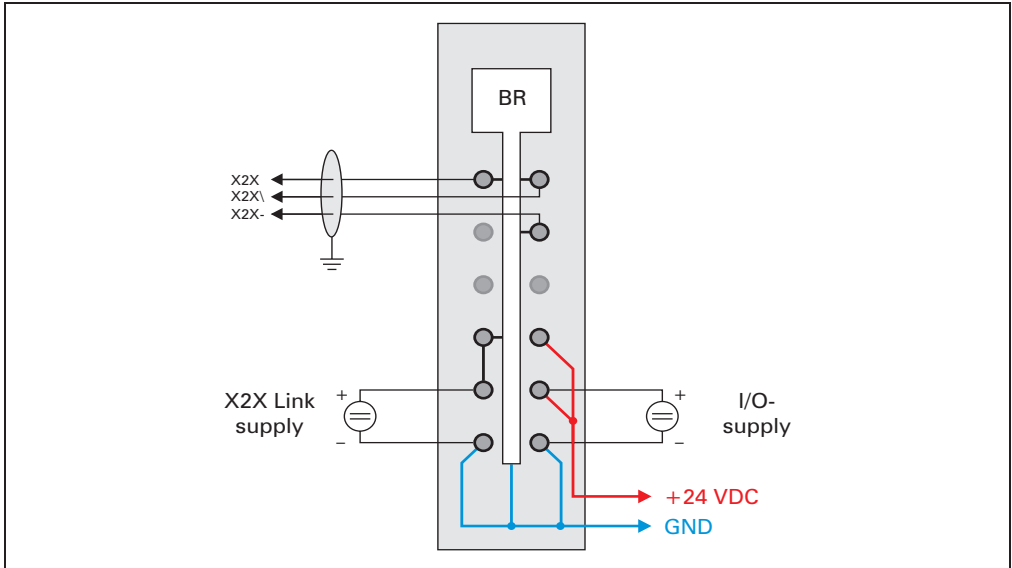


Figure 44: BR9300 - connection example with two separate supplies

With a supply and jumper

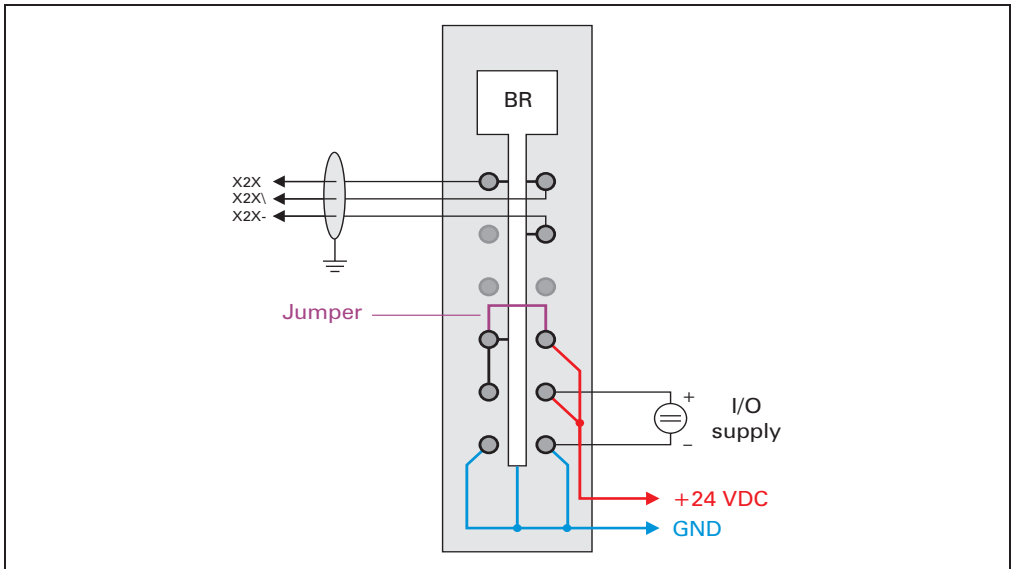


Figure 45: BR9300 - connection example with a supply and jumper

6.2.8 Module status

The following module states are monitored:

- Bus supply current
- Bus supply voltage
- 24 VDC I/O supply voltage

Bus supply current

A bus supply current of >2.3 A is displayed as a warning.

Bus supply voltage

A bus supply voltage of <4.7 V is displayed as a warning.

24 VDC I/O supply voltage

An I/O supply voltage of <20.4 V is displayed as a warning.

6.2.9 Bus supply current

The bus supply current is measured with a resolution of 0.1 A.

6.2.10 Bus supply voltage

The bus supply voltage is measured with a resolution of 0.1 V.

6.2.11 B&R ID code

Code for module identification (\$1BC1).

6.2.12 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
$\geq 100 \mu\text{s}$

6.2.13 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
2 ms

6.3 BT9100

6.3.1 General information

The bus transmitter BT9100 provides for the seamless expansion of the X20 system. The stations can be up to 100 m away from each other.

- X2X Link bus transmitter
- For seamless expansion of the system
- Up to 100 m segment lengths
- Feed for internal I/O supply

Information:

If the feed is being used for internal I/O supply, this potential group can not be supplied from any other module. An I/O module with bus module BM01 should be used to separate the potential group (see chapter 3 "Mechanical and electrical configuration", section 15.2 "Supply feed via bus transmitter" on page 56).

6.3.2 Order data

Model number	Short description	Image
	Bus transmitter	
X20BT9100	X20 bus transmitter (X2X Link)	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 63: BT9100 - order data

6.3.3 Technical data

Product ID	BT9100
Short description	
Bus transmitters	X2X Link bus transmitter with supply for I/O
I/O supply input	
Input voltage	24 VDC (-15 % / +20 %)
Fuse	Recommended pre-fusing max. 10 A slow-blow
I/O supply output	
Rated output voltage	24 VDC
Allowable contact load	10.0 A
General information	
Status indicators	X2X bus function, operating status, module status
Diagnostics Module run error X2X bus function	Yes, with status LED and software status Yes, with status LED
Power consumption Bus I/O internal	Typ. 0.45 W -
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 64: BT9100 - technical data

6.3.4 Additional technical data

Product ID	BT9100
I/O supply input	
Reverse polarity protection	No
I/O supply output	
Behavior at short circuit	Recommended protection
General information	
B&R ID code	\$1BC2

Table 65: BT9100 - additional technical data

6.3.5 Status LEDs

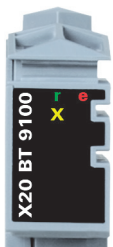
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Double flash	Indicates one of the following conditions: - I/O supply too low - The X2X bus supply is too low.
	e + r	Steady red / single green flash		Invalid firmware
	X	Orange	Off	No communication at the X2X link
			On	X2X link communication in progress

Table 66: BT9100 - status indicators

6.3.6 Pin assignments

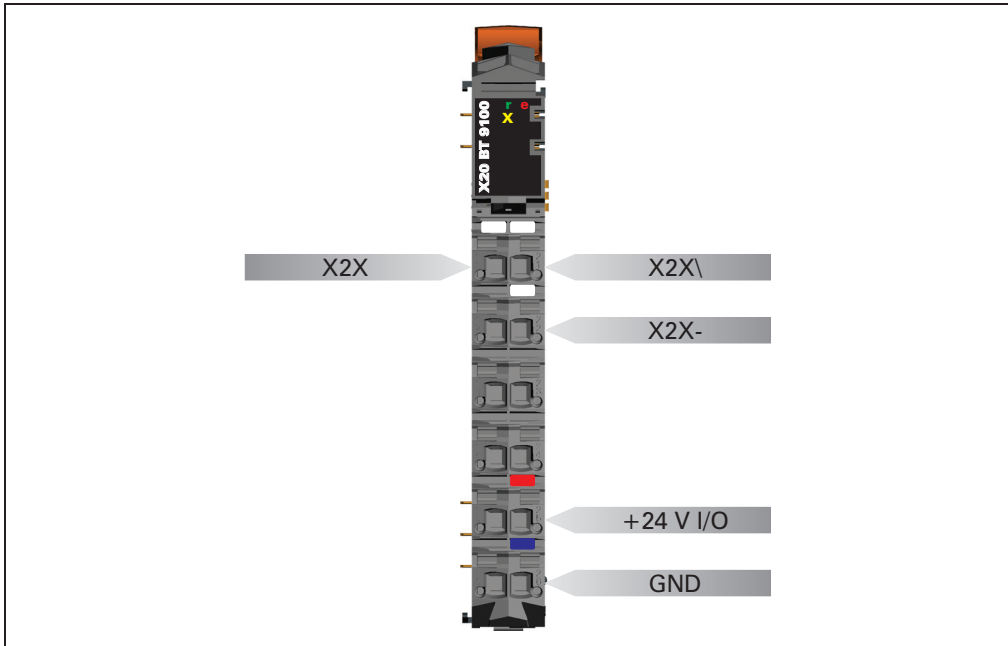


Figure 46: BT9100 - pin assignments

6.3.7 Connection examples

No feed for internal I/O supply

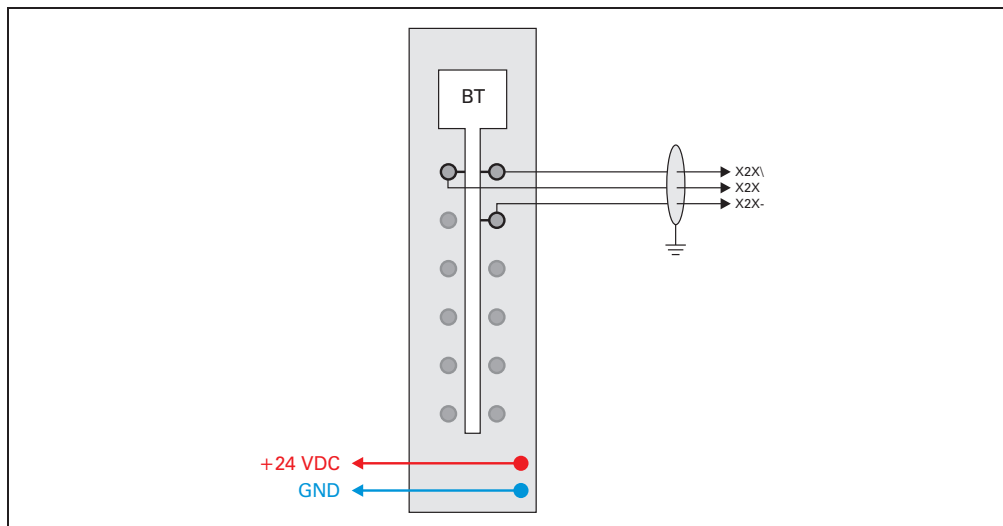


Figure 47: BT9100 - connection example - no feed for internal I/O supply

With feed for internal I/O supply

See also chapter 3 "Mechanical and electrical configuration", section 15.2 "Supply feed via bus transmitter" on page 56.

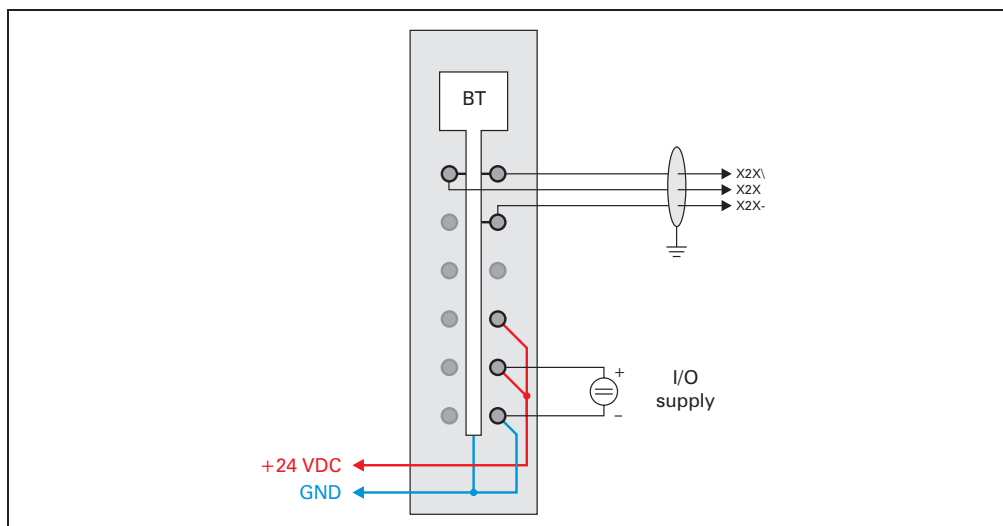


Figure 48: BT9100 - connection example - with feed for internal I/O supply

6.3.8 Connection to next X2X Link I/O node

The bus transmitter BT9100 establishes the connection to the next X2X Link based I/O node. It is important to be sure that only the data lines are connected on. X2X Link supply is system dependant.

System	X2X Link supply
X67 System	System supply PS1300
Remote I/O with X2X Link (XX modules)	24 VDC external supply
Remote valve manifold connection (XV modules)	24 VDC external supply

Table 67: BT9100 - system dependant X2X Link supply

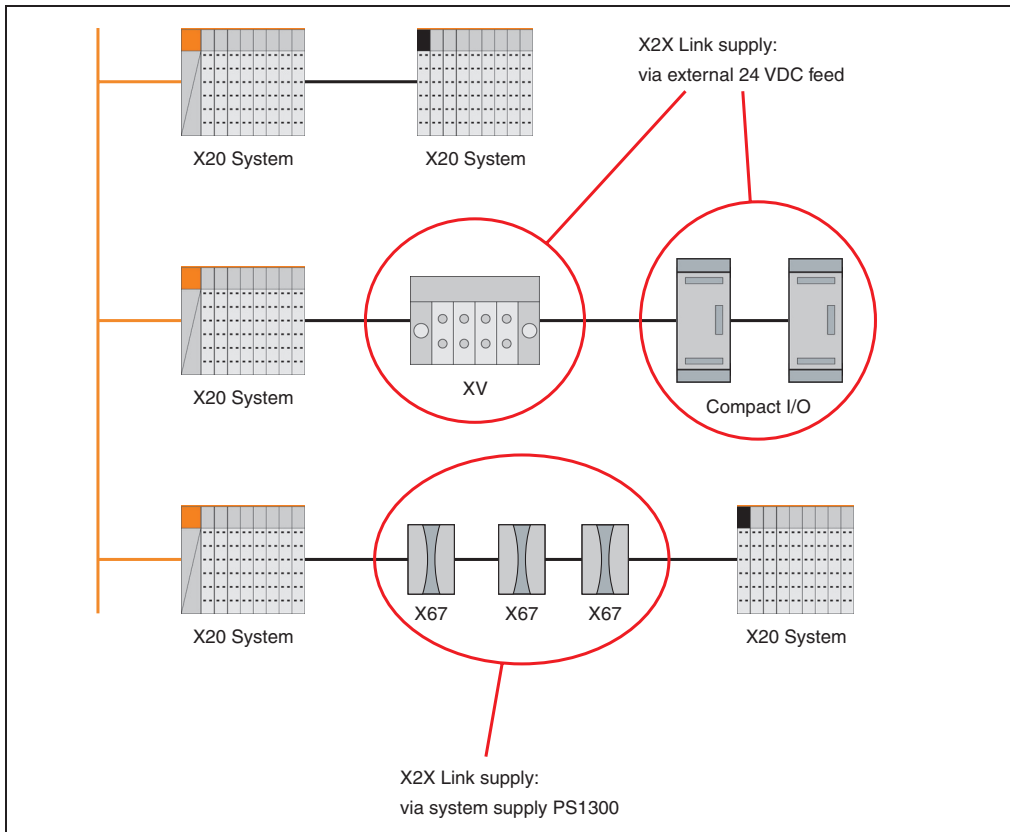


Figure 49: BT9100 - X2X Link supply according to system

6.3.9 Module status

The following module states are monitored:

- Bus supply voltage
- 24 VDC I/O supply voltage

Bus supply voltage

A bus supply voltage of <4.7 V is displayed as a warning.

24 VDC I/O supply voltage

An I/O supply voltage of <20.4 V is displayed as a warning.

6.3.10 Bus supply voltage

The bus supply voltage is measured with a resolution of 0.1 V.

6.3.11 B&R ID code

Code for module identification (\$1BC2).

6.3.12 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
$\geq 100 \mu\text{s}$

6.3.13 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
2 ms

7. Supply modules

7.1 General information

The internal I/Os and the X2X Link are fed by the supply modules. See also chapter 3 "Mechanical and electrical configuration", section 7 "X20 System infrastructure" on page 53.

7.1.1 Overview

Model number	Short description
X20PS2100	X20 supply module for internal I/O supply
X20PS3300	X20 supply module for internal I/O supply, X2X link supply

Table 68: Overview of supply modules

7.2 PS2100

7.2.1 General information

The PS2100 supply module is used for internal I/O supply.

- 24 VDC supply module for internal I/O supply

7.2.2 Order data

Model number	Short description	Image
	Supply modules	
X20PS2100	X20 supply module for internal I/O supply	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM01	X20 supply bus module, internal I/O supply is split to the left	

Table 69: PS2100 - order data

7.2.3 Technical data

Product ID	PS2100
Short description	
Power supply module	24 VDC supply module for internal I/O supply
I/O supply input	
Input voltage	24 VDC (-15 % / +20 %)
Fuse	Recommended pre-fusing max. 10 A slow-blow
I/O supply output	
Rated output voltage	24 VDC
Allowable contact load	10.0 A
General information	
Status indicators	Operating status, module status
Diagnostics Module run error	Yes, with status LED and software status
Electrical isolation I/O supply	No
Power consumption Bus I/O internal	Typ. 0.2 W -
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order supply bus module 1 x X20BM01 separately

Table 70: PS2100 - technical data

7.2.4 Additional technical data

Product ID	PS2100
I/O supply input	
Reverse polarity protection	No
I/O supply output	
Behavior at short circuit	Recommended protection
General information	
B&R ID code	\$1BBF

Table 71: PS2100 - additional technical data

7.2.5 Status LEDs

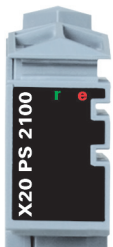
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Double flash	Indicates one of the following conditions: - I/O supply too low - The X2X bus supply is too low.
	e + r	Steady red / single green flash		Invalid firmware

Table 72: PS2100 - status indicators

7.2.6 Pin assignments

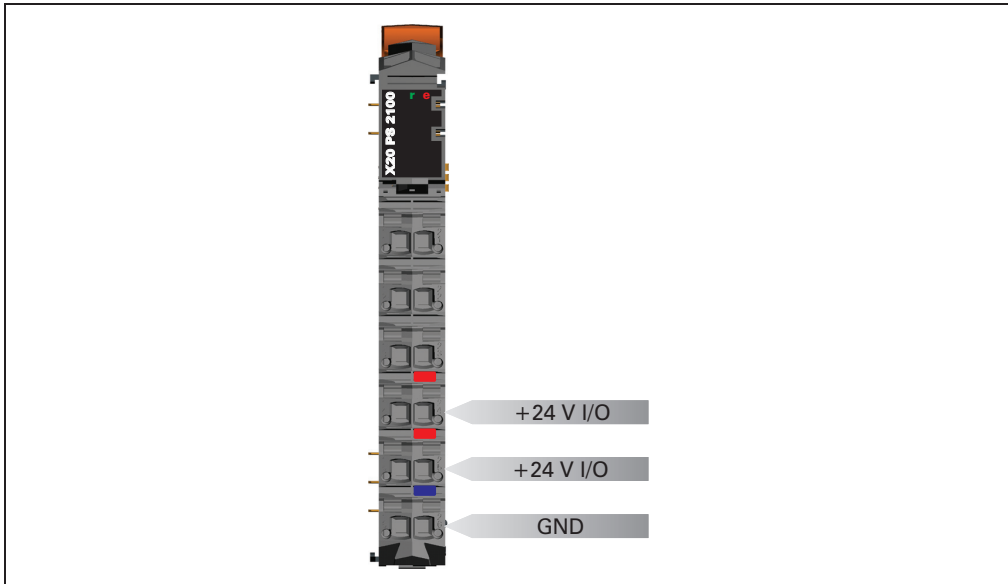


Figure 50: PS2100 - pin assignments

7.2.7 Connection example

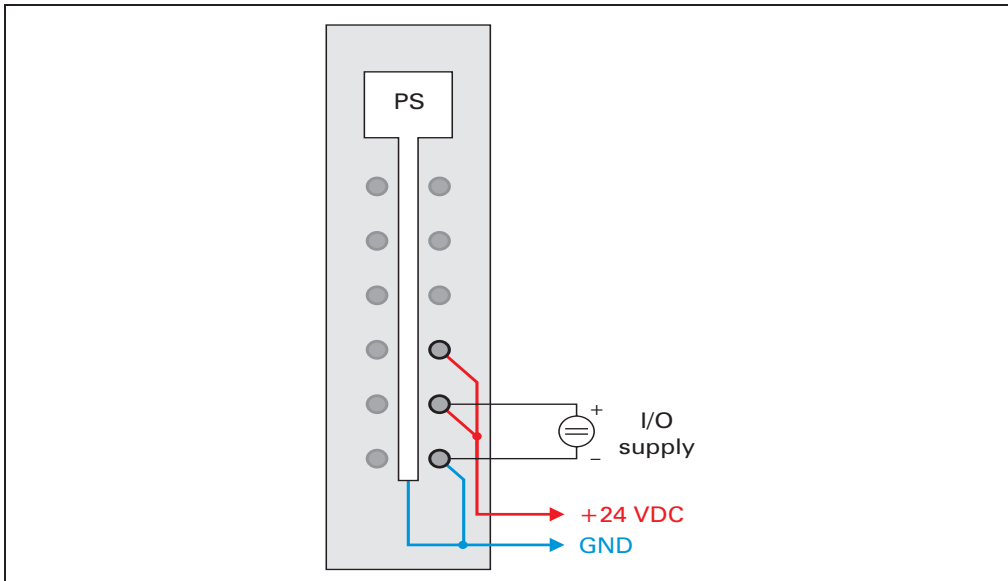


Figure 51: PS2100 - connection example

7.2.8 Module status

The following module states are monitored:

- Bus supply voltage
- 24 VDC I/O supply voltage

Bus supply voltage

A bus supply voltage of <4.7 V is displayed as a warning.

24 VDC I/O supply voltage

An I/O supply voltage of <20.4 V is displayed as a warning.

7.2.9 Bus supply voltage

The bus supply voltage is measured with a resolution of 0.1 V.

7.2.10 B&R ID code

Code for module identification (\$1BBF).

7.2.11 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
$\geq 100 \mu\text{s}$

7.2.12 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
2 ms

7.3 PS3300

7.3.1 General information

The PS3300 supply module is equipped with a feed for the X2X link as well as the internal I/O supply.

- Supply for X2X Link and internal I/O supply
- Electrical isolation of feed and X2X Link supply
- Redundancy of X2X Link supply possible by operating multiple supply modules simultaneously

7.3.2 Order data

Model number	Short description	Image
	Supply modules	
X20PS3300	X20 supply module for internal I/O supply, X2X link supply	
	Required accessories	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM01	X20 supply bus module, internal I/O supply is split to the left	

Table 73: PS3300 - order data

7.3.3 Technical data

Product ID	PS3300
Short description	
Power supply module	24 VDC supply module for I/O and bus
X2X Link supply input	
Input voltage	24 VDC (-15 % / +20 %)
Input current	Max. 0.7 A
Reverse polarity protection	Yes
Fuse	Integrated, cannot be exchanged
X2X Link supply output	
Rated output power	7.0 W
Parallel operation	Yes ¹⁾
Redundant operation	Yes
I/O supply input	
Input voltage	24 VDC (-15 % / +20 %)
Fuse	Recommended pre-fusing max. 10 A slow-blow
I/O supply output	
Rated output voltage	24 VDC
Allowable contact load	10.0 A
General information	
Status indicators	Overload, operating status, module status
Diagnostics Module run error Overload	Yes, with status LED and software status Yes, with status LED and software status
Electrical isolation X2X Link supply I/O supply	Yes No
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20

Table 74: PS3300 - technical data

Product ID	PS3300
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB12 separately Order supply bus module 1 x X20BM01 separately

Table 74: PS3300 - technical data (cont.)

- 1) In parallel operation, only 75 % of the rated power can be assumed. Please ensure that all parallel operating power supplies are switched on and off simultaneously.

7.3.4 Additional technical data

Product ID	PS3300
X2X Link supply output	
Overload behavior	Short circuit / temporary overload protection
I/O supply input	
Reverse polarity protection	No
I/O supply output	
Behavior at short circuit	Recommended protection
General information	
B&R ID code	\$1BC0

Table 75: PS3300 - additional technical data

7.3.5 Status LEDs

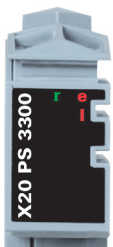
Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
			Double flash	Indicates one of the following conditions: - X2X link power supply is overloaded - I/O supply too low - Input voltage for X2X Link supply too low
	e + r	Steady red / single green flash		Invalid firmware
	l	Red	Off	X2X link supply in the acceptable range
			On	X2X link power supply is overloaded

Table 76: PS3300 - status indicators

7.3.6 Pin assignments

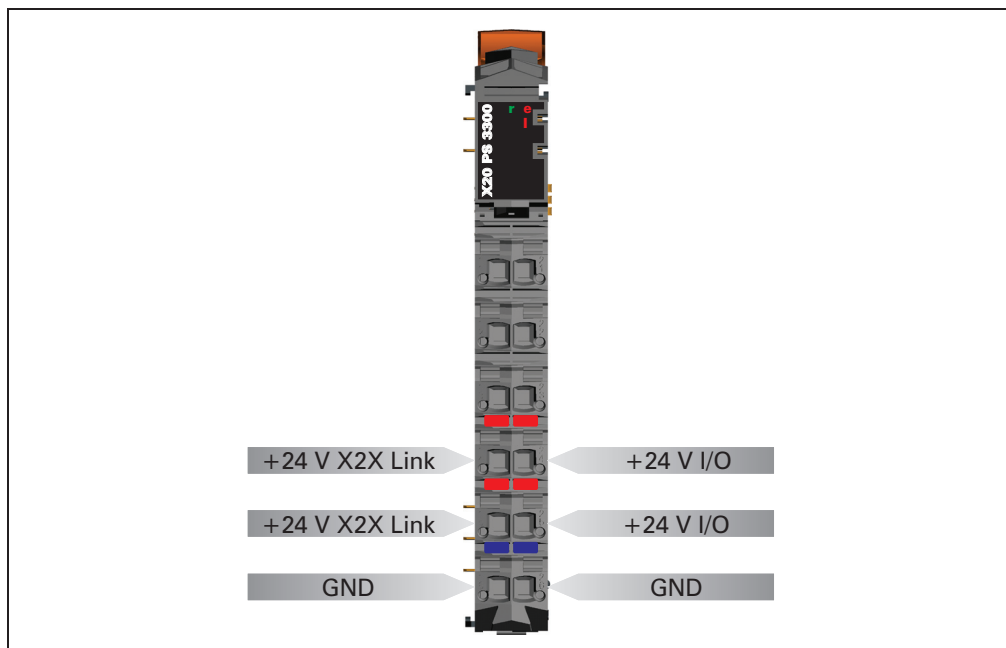


Figure 52: PS3300 - pin assignments

7.3.7 Connection examples

With two separate supplies

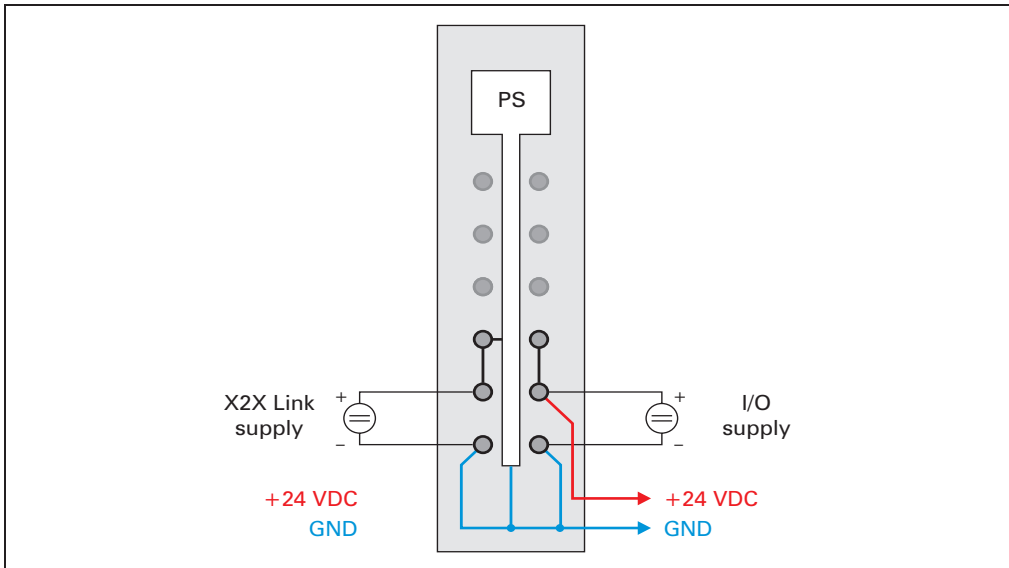


Figure 53: PS3300 - connection example with two separate supplies

With a supply and jumper

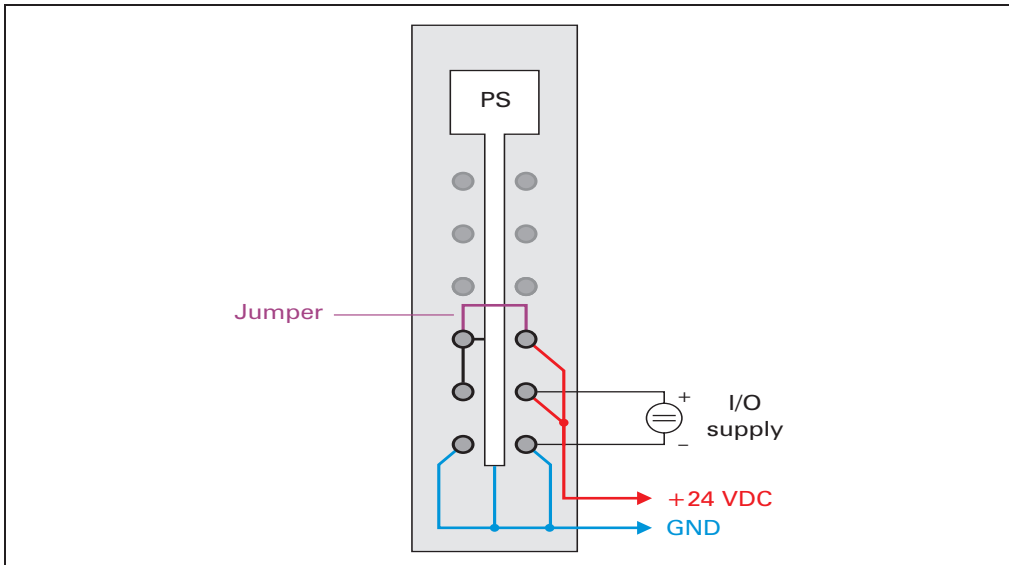


Figure 54: PS3300 - connection example with a supply and jumper

7.3.8 Module status

The following module states are monitored:

- Bus supply current
- Bus supply voltage
- 24 VDC I/O supply voltage

Bus supply current

A bus supply current of >2.3 A is displayed as a warning.

Bus supply voltage

A bus supply voltage of <4.7 V is displayed as a warning.

24 VDC I/O supply voltage

An I/O supply voltage of <20.4 V is displayed as a warning.

7.3.9 Bus supply current

The bus supply current is measured with a resolution of 0.1 A.

7.3.10 Bus supply voltage

The bus supply voltage is measured with a resolution of 0.1 V.

7.3.11 B&R ID code

Code for module identification (\$1BC0).

7.3.12 Minimum cycle time

The minimum cycle time is the minimum time needed for the bus cycle to be shut down without a communication error occurring. It should be noted that very fast cycles have reduced the idle time needed for handling monitoring, diagnostics and acyclic commands.

Minimum cycle time
$\geq 100 \mu\text{s}$

7.3.13 Minimum I/O update time

The minimum I/O update time refers to the minimum time it takes for the bus cycle to shut down, so that in each cycle an I/O update takes place.

Minimum I/O update time
2 ms

8. Digital input modules

8.1 General information

Digital input modules convert binary process signals into the internal signal level required by the PLC. The states of the digital inputs are indicated with status LEDs.

8.1.1 Input filter

An input filter is available for each input. The input delay is shown in the technical data sections. Disturbance pulses which are shorter than the input delay are suppressed by the input filter.

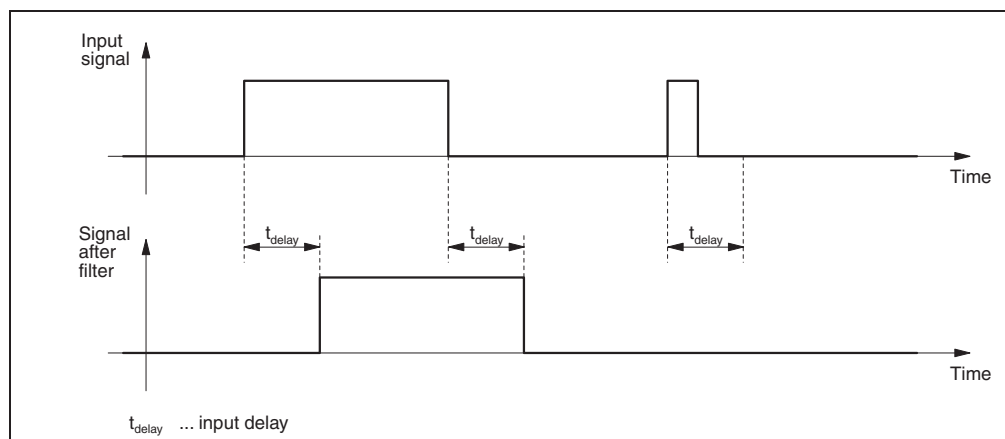


Figure 55: Input filter

8.1.2 Overview

Module	DI2371	DI4371	DI6371	DI6372	DI9371	DI9372
Number of inputs	2	4	6	6	12	12
Rated voltage	24 VDC					
Input filter Hardware Software	$\leq 100 \mu\text{s}$ Default 1 ms, can be configured between 0 and 25 ms in 0.2 ms intervals					
Input circuit	Sink	Sink	Sink	Source	Sink	Source
Connection type	3-wire connections	3-wire connections	2-wire connections	2-wire connections	1-wire connections	1-wire connections

Table 77: Digital input module overview

Module	DI2377
Number of inputs	2
Rated voltage	24 VDC
Input filter Hardware Software	$\leq 10 \mu\text{s}$ Default 0 ms, can be configured between 0 and 25 ms in 0.2 ms intervals
Input circuit	Sink
Connection type	3-wire connections
Additional functions for inputs	Event counting, gate measurement

Table 78: Digital input module with additional functions

8.2 DI2371

8.2.1 General information

The DI2371 module is equipped with two inputs for 3-wire connections. It is designed for standard X20 6-pin terminal blocks. If needed (e.g. for logistical reasons), the 12-pin terminal block can also be used.

- 2 digital inputs
- Sink connection
- 3-wire connection
- 24 VDC and GND for sensor supply
- Software input filter can be configured for entire module

8.2.2 Order data

Model number	Short description	Image
	Digital input module	
X20DI2371	X20 digital input module, 2 inputs, 24 VDC, sink, configurable input filter, 3-wire connections	
	Required accessories	
X20TB06	Standard X20 terminal block (6-pin)	
X20TB12	Standard X20 terminal block (12-pin)	
X20BM11	X20 standard bus module, internal I/O supply is interconnected	

Table 79: DI2371 - order data

8.2.3 Technical data

Product ID	DI2371
Short description	
I/O module	Two 24 VDC digital inputs for 3-wire connections
Digital inputs	
Rated voltage	24 VDC
Input filter Hardware Software	≤100 µs Default 1 ms, can be configured between 0 and 25 ms in 0.2 ms intervals
Input circuit	Sink
Sensor supply	0.5 A total current
General information	
Status indicators	I/O function per channel, operating state, module status
Diagnostics Module run error	Yes, with status LED and software status
Electrical isolation Channel - Bus Channel - Channel	Yes No
Power consumption Bus I/O internal	Typ. 0.12 W Max. 0.26 W
Certification	CE, C-UL-US, GOST-R (in development)
Operational conditions	
Operating temperature Horizontal installation Vertical installation	0 °C to +55 °C 0 °C to +50 °C
Humidity	5 to 95%, non-condensing
Mounting orientation	Horizontal or vertical
Installation at altitudes above sea level 0 - 2000 m >2000 m	No derating Reduction of environmental temperature by 0.5 °C per 100 m
Protection type	IP20
Storage and transport conditions	
Temperature	-25 °C to +70 °C
Humidity	5 to 95%, non-condensing
Mechanical characteristics	
Spacing	12.5 +0.2 mm
Note	Order standard terminal block 1 x X20TB06 or X20TB12 separately Order standard bus module 1 x X20BM11 separately

Table 80: DI2371 - technical data

8.2.4 Additional technical data

Product ID	DI2371
Sensor supply	
Voltage	Module supply minus voltage drop for short circuit protection
Voltage drop for short circuit protection at 500 mA	Max. 2 VDC
Short circuit protection	Yes
Power consumption Sensor supply	Max. 12.0 W ¹⁾
Digital inputs	
Input voltage	24 VDC (-15 % / +20 %)
Input current at 24 VDC	Typ. 3.75 mA
Input resistance	Typ. 6.4 kΩ
Switching threshold Low High	<5 VDC >15 VDC
Isolation voltage betw. channel and bus	500 V _{eff}
General information	
B&R ID code	\$1B8D

Table 81: DI2371 - additional technical data

1) The power consumption of the sensors connected to the module may not exceed 12 W.

8.2.5 Status LEDs

Image	LED	Color	Status	Description
	r	Green	Off	Module not supplied
			Single flash	Reset mode
			Blinking	Preoperational mode
			On	RUN mode
	e	Red	Off	Module not supplied or everything is OK
	e + r	Steady red / single green flash		Invalid firmware
	1 - 2	Green		Input status of the corresponding digital input

Table 82: DI2371 - status indicators